FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

E- COMMERCE ORDER MANAGEMENT SYSTEM (OMS)

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April 2026 | Version 1.0

Document Control

Version	Date	Author	Description
0.1	April 2026	Revathi Kamindla	Initial Draft
1.0	April 2026	Revathi Kamindla	Final Version for Review

Approval Sign-Off

Role	Name	Signature	Date
Business Owner			
Technical Lead			
Business Analyst	Revathi Kamindla		April 2026

1. Introduction

This Functional Requirements Document (FRD) defines the detailed system-level functionalities of the E-Commerce Order Management System (OMS). It translates the high-level business requirements captured in the BRD into specific, testable, and implementable system behaviours.

This document is intended for developers, testers, and technical architects involved in building and validating the OMS.

2. System Overview

The OMS is a centralized web-based platform that manages the complete order lifecycle from order placement to delivery and returns. It provides:

- A customer-facing portal for browsing, ordering, tracking, and returns
- An admin dashboard for order monitoring and management
- Automated notifications and status updates throughout the order lifecycle
- Integration touchpoints with payment gateways and delivery partners

3. User Roles and Permissions

Role	Access Level	Key Permissions
Customer	Standard	Register, login, browse, order, track, cancel, return
Administrator	Elevated	View all orders, update status, manage cancellations and returns
Vendor	Limited	View assigned orders, update packing status
Delivery Partner	Limited	Update shipment and delivery status
Super Admin	Full	All system access including user management

4. Functional Requirements

4.1 User Management

FR ID	Functional Requirement	Linked BR	Priority
FR-01	System shall allow users to register with valid personal details	BR-01	Must Have
FR-02	System shall authenticate users through secure login credentials	BR-01	Must Have
FR-03	System shall allow users to update their profile information	BR-01	Should Have
FR-04	System shall allow users to reset their password securely	BR-01	Must Have
FR-05	System shall automatically log out inactive users after 15 minutes	BR-01	Should Have

4.2 Product and Order Management

FR ID	Functional Requirement	Linked BR	Priority
FR-06	System shall allow users to browse available products	BR-02	Must Have
FR-07	System shall allow users to add products to the cart	BR-02	Must Have
FR-08	System shall allow users to place an order successfully	BR-02	Must Have
FR-09	System shall generate a unique Order ID for each transaction	BR-03	Must Have
FR-10	System shall display order confirmation after successful placement	BR-02	Must Have
FR-11	System shall save cart items if session expires before checkout	BR-02	Could Have

4.3 Order Tracking and Status Updates

FR ID	Functional Requirement	Linked BR	Priority
FR-12	System shall provide real-time order status updates	BR-04	Must Have
FR-13	System shall update order status at each stage of the lifecycle	BR-04	Must Have
FR-14	System shall allow users to view complete order history	BR-04	Should Have
FR-15	System shall display estimated delivery date after shipment	BR-04	Should Have
FR-16	System shall notify users of key order status changes	BR-08	Should Have

4.4 Payment Management

FR ID	Functional Requirement	Linked BR	Priority
FR-17	System shall support payment processing through integrated gateway	BR-02	Must Have
FR-18	System shall update payment status (Success, Pending, Failed)	BR-02	Must Have
FR-19	System shall generate payment confirmation upon successful transaction	BR-02	Must Have
FR-20	System shall allow customer to retry payment upon failure	BR-02	Must Have

4.5 Order Cancellation and Returns

FR ID	Functional Requirement	Linked BR	Priority
FR-21	System shall allow users to cancel orders before shipment	BR-05	Must Have
FR-22	System shall validate cancellation eligibility based on order status	BR-05	Must Have
FR-23	System shall update order status to "Cancelled" upon successful cancellation	BR-05	Must Have
FR-24	System shall allow users to initiate return requests after delivery	BR-06	Should Have
FR-25	System shall process refund requests based on defined business rules	BR-06	Should Have
FR-26	System shall notify customer of cancellation or return confirmation	BR-08	Should Have

4.6 Admin Functionalities

FR ID	Functional Requirement	Linked BR	Priority
FR-27	System shall allow administrators to view all customer orders	BR-07	Must Have
FR-28	System shall allow administrators to update order status manually	BR-07	Must Have
FR-29	System shall allow administrators to filter orders by status and date	BR-07	Should Have
FR-30	System shall allow administrators to manage cancellations and returns	BR-07	Must Have
FR-31	System shall provide a dashboard overview of all order activities	BR-09	Should Have

5. Non-Functional Requirements

5.1 Performance

NFR ID	Requirement	Standard	Linked BR
NFR-P01	Page load time	≤ 3 seconds	NFR-01
NFR-P02	Order details retrieval	≤ 2 seconds	NFR-01
NFR-P03	Concurrent user support	10,000 users simultaneously	NFR-02
NFR-P04	Order status update	Real-time within 5 seconds	NFR-01

5.2 Security

NFR ID	Requirement	Standard	Linked BR
NFR-S01	Password encryption	bcrypt or equivalent	NFR-04
NFR-S02	Data encryption	AES-256 or equivalent	NFR-04
NFR-S03	Role-based access control	RBAC implementation	NFR-04
NFR-S04	Session management	Auto logout after 15 mins	NFR-04

5.3 Availability

NFR ID	Requirement	Standard	Linked BR
NFR-A01	System uptime	99.9% SLA	NFR-03
NFR-A02	Recovery Time Objective	RTO ≤ 15 minutes	NFR-05
NFR-A03	Recovery Point Objective	RPO ≤ 1 hour	NFR-06

5.4 Scalability

NFR ID	Requirement	Standard	Linked BR
NFR-SC01	Order volume at peak	1,00,000 orders/day	NFR-07
NFR-SC02	Annual user growth	30% growth without redesign	NFR-07

6. User Stories with Acceptance Criteria

User Stories represent system requirements from the end-user perspective in an Agile format. They help in understanding user needs and expected business value in a simple and structured way.

Each user story follows the standard Agile format:
As a, I want, so that.

6.1 Customer User Stories

User Story -01: User Registration and Login

As a customer, I want to register and log in to the system so that I can access my account securely.

Acceptance Criteria	
AC-01	Given the customer provides valid details, when they register, then account is created successfully
AC-02	Given the customer enters correct credentials, when they log in, then they are redirected to the home page
AC-03	Given the customer enters incorrect credentials, when they attempt login, then an error message is displayed
AC-04	Given the customer is inactive for 15 minutes, when the session expires, then they are automatically logged out

User Story -02: Browse Products

As a customer, I want to browse products so that I can select items for purchase.

Acceptance Criteria	
AC-01	Given the customer is logged in, when they navigate to the product catalogue, then all available products are displayed
AC-02	Given a product is out of stock, when the customer views it, then an "Out of Stock" label is displayed
AC-03	Given the customer searches for a product, when results load, then relevant products appear within 2 seconds

User Story -03: Place Order

As a customer, I want to place an order so that I can purchase selected products.

Acceptance Criteria	
AC-01	Given the customer completes payment, when the order is confirmed, then a unique Order ID is generated
AC-02	Given payment fails, when the customer is notified, then the order is not placed and they are prompted to retry
AC-03	Given the order is placed successfully, when confirmation is generated, then the customer receives a system notification

User Story -04: Track Order

As a customer, I want to track my order so that I can know the delivery status in real time.

Acceptance Criteria	
AC-01	Given the order is placed, when the customer navigates to "My Orders", then current order status is displayed
AC-02	Given the order is shipped, when the customer views tracking, then tracking ID and estimated delivery date are shown
AC-03	Given the order is delivered, when the customer checks status, then "Delivered" is shown with date and timestamp

User Story -05: Cancel Order

As a customer, I want to cancel my order so that I can avoid unwanted purchases.

Acceptance Criteria	
AC-01	Given the order is not yet shipped, when the customer clicks "Cancel Order", then the order status is updated to "Cancelled"
AC-02	Given the order has already been shipped, when the customer attempts cancellation, then the system displays "Cancellation not available"
AC-03	Given the order is cancelled, when the process is complete, then the customer receives a cancellation confirmation notification

User Story -06: Initiate Return

As a customer, I want to initiate a return so that I can return faulty or unwanted products.

Acceptance Criteria	
AC-01	Given the order is delivered, when the customer selects "Return Order", then a return request form is displayed
AC-02	Given the return request is submitted, when it is validated, then status is updated to "Return Initiated"
AC-03	Given the return is not eligible (e.g., return window expired), when the customer attempts return, then an appropriate message is shown

6.2 Admin User Stories

User Story -07: View All Orders

As an admin, I want to view all customer orders so that I can monitor system activity.

Acceptance Criteria	
AC-01	Given the admin is logged in, when they navigate to the dashboard, then all orders with statuses are displayed
AC-02	Given the admin applies a filter, when results load, then only orders matching the filter criteria are shown

User Story -08: Update Order Status

As an admin, I want to update order status so that customers receive accurate tracking information.

Acceptance Criteria	
AC-01	Given the admin selects an order, when they update the status, then the new status is reflected immediately in the system
AC-02	Given the status is updated, when the change is saved, then the customer receives an automated notification

User Story -09: Manage Cancellations and Returns

As an admin, I want to manage cancellations and returns so that business rules are followed.

Acceptance Criteria	
AC-01	Given a cancellation request is raised, when the admin reviews it, then they can approve or reject based on eligibility
AC-02	Given a return is approved, when the admin confirms it, then the order status updates to "Return Approved"

6.3 Business Value of User Stories

- Provides clear understanding of user needs and expectations
- Helps in breaking down requirements into actionable tasks for development
- Improves communication between business and technical teams
- Supports Agile development and sprint planning
- Ensures user-centric system design and implementation

7. Use Cases

Use Cases describe the interaction between users (actors) and the system to achieve specific goals. They provide a detailed view of how different functionalities of the E-commerce Order Management System (OMS) are executed in real-world scenarios.

7.1 Use Case: Place Order

Field	Details
Use Case ID	UC-01
Use Case Name	Place Order
Actor	Customer
Description	Customer places an order for selected products
Preconditions	Customer is logged in; products are available in catalogue
Postconditions	Order is successfully created with a unique Order ID

Main Flow:

1. Customer selects products and adds to cart
2. Customer proceeds to checkout
3. System calculates total order amount
4. Customer selects payment method and completes payment
5. System validates payment confirmation
6. System generates unique Order ID
7. Order confirmation is displayed to customer
8. System notification is sent to customer

Alternate Flow — Payment Failure:

- Step 5a: Payment gateway returns failure response
- System displays "Payment Failed" message
- Customer is prompted to retry or use alternate payment method
- Order remains in "Pending Payment" status

Exception Flow — Session Timeout:

- If customer is inactive for 15 minutes during checkout
- System automatically logs out the customer
- Cart items are saved for the next login session

7.2 Use Case: Track Order

Field	Details
Use Case ID	UC-02
Use Case Name	Track Order
Actor	Customer
Description	Customer views real-time status of their placed order
Preconditions	Order has been placed successfully
Postconditions	Customer is informed of current order status

Main Flow:

1. Customer logs into system
2. Customer navigates to "My Orders" section
3. System retrieves and displays order list
4. Customer selects specific order
5. System displays current order stage with timestamp

Alternate Flow — No Orders Found:

- System displays "No orders found" message
- Customer is prompted to start shopping

7.3 Use Case: Cancel Order

Field	Details
Use Case ID	UC-03
Use Case Name	Cancel Order

Actor	Customer
Description	Customer cancels an order that has not yet been shipped
Preconditions	Order exists and has not been shipped
Postconditions	Order status updated to "Cancelled"

Main Flow:

1. Customer navigates to "My Orders"
2. Customer selects the order to cancel
3. Customer clicks "Cancel Order"
4. System validates cancellation eligibility
5. System cancels the order and updates status to "Cancelled"
6. Customer receives cancellation notification

Alternate Flow — Order Already Shipped:

- System detects order is in "Shipped" or later status
- System displays "Cancellation not available at this stage"
- Customer is informed to raise a return request after delivery

7.4 Use Case: Process Order (Admin)

Field	Details
Use Case ID	UC-04
Use Case Name	Process Order
Actor	Admin
Description	Admin reviews and processes confirmed orders
Preconditions	Order is placed and payment is confirmed
Postconditions	Order status updated to "Processed" and moves to packaging stage

Main Flow:

1. Admin logs into system
2. Admin navigates to pending orders on dashboard
3. Admin reviews order details
4. Admin processes the order
5. System updates order status to "Processed"
6. Order is passed to vendor for packaging

Exception Flow — Incomplete Order Information:

- Admin identifies missing or incorrect order information
- Admin flags the order and contacts customer for clarification
- Order remains in "On Hold" status until resolved

7.5 Business Value of Use Cases

- Provides clear step-by-step system interaction view
- Helps bridge gap between business requirements and technical implementation
- Improves clarity for developers and testers
- Supports system design and test case creation
- Enhances understanding of real-world system behaviour

8. Process Flow (Order Lifecycle)

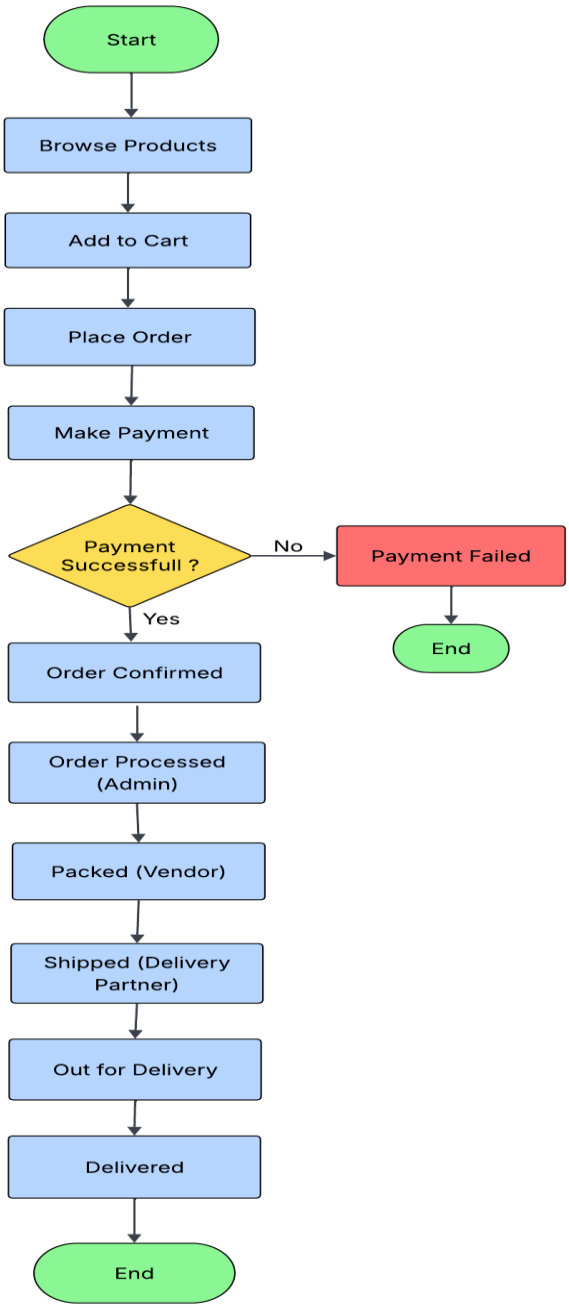
The Process Flow defines the end-to-end sequence of activities involved in the E-commerce Order Management System (OMS). It represents how an order moves through different stages from initiation to completion, ensuring clarity in business operations and stakeholder responsibilities.

8.1 Order Lifecycle Flow

The typical order process in the system follows the steps below:

1. Customer logs into the system
2. Customer browses and selects products
3. Customer adds products to cart and places an order
4. System generates a unique Order ID
5. Payment is processed through the payment gateway
6. Order confirmation is generated upon successful payment
7. Order is processed by the system/admin
8. Order is packed by vendor/seller
9. Order is shipped by delivery partner
10. Customer receives the order
11. Order status is updated to “Delivered”
12. Customer may initiate return or refund (if applicable)

8.5 Process Flow Diagram



"This flow represents the end-to-end order lifecycle from order placement to delivery and completion."

9. Requirements Traceability Matrix (RTM)

BR ID	Business Requirement	FR ID	Functional Requirement	User Story	Use Case	Test Case
BR-01	User registration and login	FR-01, FR-02	Register and authenticate users	US-01	—	TC-001, TC-002
BR-02	Product browsing and order placement	FR-06, FR-07, FR-08	Browse, add to cart, place order	US-02, US-03	UC-01	TC-003, TC-004
BR-03	Unique Order ID generation	FR-09	Generate unique Order ID	US-03	UC-01	TC-003
BR-04	Real-time order tracking	FR-12, FR-13	Real-time status updates	US-04	UC-02	TC-005
BR-05	Order cancellation	FR-21, FR-22, FR-23	Cancel before shipment	US-05	UC-03	TC-006, TC-007
BR-06	Return and refund	FR-24, FR-25	Return after delivery	US-06	—	TC-008
BR-07	Admin order management	FR-27, FR-28, FR-30	Admin views and updates orders	US-07, US-08, US-09	UC-04	TC-009, TC-010
BR-08	Customer notifications	FR-16, FR-26	Notify users of status changes	US-03, US-05	UC-01, UC-03	TC-008

10. UAT Test Scenarios

Test Case ID	Linked User Story	Test Scenario	Steps	Expected Result
TC-001	US-01	Valid user registration	Enter valid details → Submit → Check confirmation	Account created successfully
TC-002	US-01	Invalid login attempt	Enter wrong password → Click Login	Error message displayed
TC-003	US-03	Successful order placement	Add to cart → Checkout → Pay → Confirm	Unique Order ID generated
TC-004	US-03	Payment failure during checkout	Enter invalid card → Submit payment	Error shown; order not placed
TC-005	US-04	Order tracking after shipment	Login → My Orders → Select order	Tracking ID and delivery date shown
TC-006	US-05	Cancel order before shipment	Select order → Cancel → Confirm	Status updated to "Cancelled"
TC-007	US-05	Cancel order after shipment	Select shipped order → Attempt cancel	"Cancellation not available" message shown
TC-008	US-06	Return request after delivery	Select delivered order → Initiate return → Submit	Status updated to "Return Initiated"
TC-009	US-08	Admin updates order status	Admin login → Select order → Update status	Status updated; customer notified
TC-010	US-09	Admin approves return request	Admin reviews return → Approves	Status updated to "Return Approved"

11. Approval

Role	Name	Approval Status	Date
Business Owner		Pending	
Technical Lead		Pending	
Business Analyst	Revathi Kamindla	Approved	April 2026